Vector Calculus and Complex Analysis Homework 8

1. Consider the function $f(z) = x^3 + 3xy^2 + (3x^2y + y^3)i,\ z = x + iy.$ Find the region where

(a) the Cauchy–Riemann relations are satisfied;

(b) the function is differentiable;

(c) the function is analytic.

Express $f'(z)$ in terms of x and y if f is differentiable.

Solution

(a) $u = x^3 + 3xy^2, \quad v = 3x^2y + y^3$

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 3x^2 + 3y^2, \quad 6xy = \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -6xy \quad \Rightarrow \quad x = 0 \quad \text{or} \quad y = 0$

f satisfies C-R equations when $x = 0 \quad \text{or} \quad y = 0.$

(b) All 1-st partial derivatives are continuous. f is differentiable when $x = 0 \quad \text{or} \quad y = 0.$

(c) f is nowhere analytic because $\{(x, y) : x = 0 \quad \text{or} \quad y = 0\}$ is not an open connected set.

$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = 3x^2 + 3y^2 + 6ixy \qquad \text{on } x = 0 \quad \text{or} \quad y = 0$

2. (a) By considering $3 - 4i = (u + iv)^2,$ where u and v are real. Show that $(3 - 4i)^{1/2} = \pm(2 - i).$

(b) By considering $3 - 4i = r(\cos\theta + i\sin\theta) = re^{i\theta}.$ Since $\sin^2\frac{\theta}{2} = \frac{1 - \cos\theta}{2}$ and

$\cos^2\frac{\theta}{2} = \frac{1 + \cos\theta}{2},$ we have

$$(3 - 4i)^{1/2} = \pm\sqrt{r}e^{i\theta/2} = \pm\sqrt{r}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right) = \pm\sqrt{r}\left(\sqrt{\frac{1 + \cos\theta}{2}} + i\sqrt{\frac{1 - \cos\theta}{2}}\right).$$

Show that $(3 - 4i)^{1/2} = \pm(2 + i).$

(c) By comparing the result of (a) and (b), which part goes wrong?

Solution

(a) $3-4i=(u+iv)^2=u^2-v^2+2iuv$

$$\begin{cases} 3=u^2-v^2 \\ -4=2uv \end{cases}$$

$3u^2=u^4-u^2v^2=u^4-4$

$u^4-3u^2-4=0$

$(u^2-4)(u^2+1)=0$

$u=\pm 2, \quad v=\mp 1$

$(3-4i)^{1/2}=\pm(2-i).$

(b) $r=5, \quad \cos\theta=0.6,$

$$\begin{aligned}(3-4i)^{1/2} &= \pm\sqrt{r}\left(\sqrt{\frac{1+\cos\theta}{2}}+i\sqrt{\frac{1-\cos\theta}{2}}\right) \\ &= \pm\sqrt{5}\left(\sqrt{\frac{1+0.6}{2}}+i\sqrt{\frac{1-0.6}{2}}\right) \\ &= \pm\sqrt{5}(\sqrt{0.8}+i\sqrt{0.2}) \\ &= \pm(2+i).\end{aligned}$$

(c) $$\left(r(\cos\theta+i\sin\theta)\right)^{1/2}=\begin{cases} \pm\sqrt{r}\left(\sqrt{\frac{1+\cos\theta}{2}}+i\sqrt{\frac{1-\cos\theta}{2}}\right) & \text{if } 0\le\theta\le\pi \\ \pm\sqrt{r}\left(\sqrt{\frac{1+\cos\theta}{2}}-i\sqrt{\frac{1-\cos\theta}{2}}\right) & \text{if } -\pi<\theta<0 \end{cases}$$

because $\tan\frac{\theta}{2}\ge 0$ if $0\le\theta\le\pi$ and $\tan\frac{\theta}{2}<0$ if $-\pi<\theta<0.$

3. Solve $\sin z=\dfrac{1}{\sin 2\theta}$ where $0<\theta<\frac{\pi}{2}$.

Solution

$$\sin z = \frac{1}{\sin 2\theta}$$

$$\frac{e^{iz} - e^{-iz}}{2i} = \frac{1}{\sin 2\theta}$$

$$e^{iz}\sin 2\theta - 2i - e^{-iz}\sin 2\theta = 0$$

$$e^{2iz}\sin 2\theta - 2ie^{iz} - \sin 2\theta = 0$$

$$e^{iz} = \frac{2i \pm \sqrt{(-2i)^2 + 4\sin^2 2\theta}}{2\sin 2\theta} = \frac{1 \pm \cos 2\theta}{\sin 2\theta} i = \frac{2\cos^2\theta}{2\sin\theta\cos\theta} i \quad \text{or} \quad \frac{2\sin^2\theta}{2\sin\theta\cos\theta} i$$

$$= i\cot\theta \quad \text{or} \quad i\tan\theta$$

$$iz = \pm \text{Log}(\tan\theta) + \left(2k + \frac{1}{2}\right) i\pi$$

$$z = \left(2k + \frac{1}{2}\right)\pi \pm i\text{Log}(\tan\theta)$$

4. (a) Suppose f is analytic and real-valued in a domain D. Show that f is constant in D.

(b) Suppose f and $\overline{f}$ are analytic in a domain D. Show that f is constant in D.

(c) Suppose f is analytic in a domain D and $|f|$ is constant in D. Show that f is constant in D.

Solution

(a) $\text{Im}\, f(z) = 0$ in $D \Longrightarrow \dfrac{\partial v}{\partial x} = \dfrac{\partial v}{\partial y} = 0 \Longrightarrow \dfrac{\partial u}{\partial x} = 0$

$\Longrightarrow f'(z) = \dfrac{\partial u}{\partial x} + i\dfrac{\partial v}{\partial x} = 0 \Longrightarrow f$ is constant in D.

(b) $\text{Re}\, f(z) = \dfrac{1}{2}[f(z) + \overline{f(z)}]$ is real valued and analytic if both f and $\overline{f}$ are analytic. Hence $\text{Re}\, f(z)$ is constant by (a). It follows that $f(z)$ is constant.

(c) $|f(z)|$ constant in $D \Longrightarrow |f(z)|^2 = u^2 + v^2$ is constant in D. If $u = 0$ or $v = 0$ in D, then f is constant. Otherwise,

$$\frac{\partial |f|^2}{\partial x} = 2u\frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x} = 0$$
$$\frac{\partial |f|^2}{\partial y} = 2u\frac{\partial u}{\partial y} + 2v\frac{\partial v}{\partial y} = -2u\frac{\partial v}{\partial x} + 2v\frac{\partial u}{\partial x} = 0$$

$$\Longrightarrow \frac{1}{2}v\frac{\partial |f|^2}{\partial x} - \frac{1}{2}u\frac{\partial |f|^2}{\partial y} = 0 = (u^2 + v^2)\frac{\partial v}{\partial x}$$
$$\Longrightarrow \frac{\partial v}{\partial x} = 0 \Longrightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial u}{\partial y} = 0$$
$$\Longrightarrow f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = 0$$
$$\Longrightarrow f \text{ is constant in } D.$$

5. Show that the following functions are harmonic, and find its harmonic conjugates:

(a) $xy + 3x^2y - y^3$

(b) $e^{x^2-y^2}\cos(2xy)$

(c) $\tan^{-1}(y/x),\ x > 0$

(d) $\dfrac{x}{x^2 + y^2}$

Solution

(a) Set

$$u(x,y) = xy+3x^2y-y^3 \Rightarrow \begin{cases} u_x^{'} = y+6xy \\ u_y^{'} = x+3x^2-3y^2 \end{cases} \Rightarrow \begin{cases} u_{xx}^{''} = 6y \\ u_{yy}^{'} = -6y \end{cases} \Rightarrow \triangle u = 0,$$

thus is the function $u(x,y)$ is harmonic. Cauchy Riemann give us

$$\begin{cases} v_x^{'} = -u_y^{'} = -x-3x^2+3y^2 \Rightarrow v(x,y) = -\frac{x^2}{2} - x^3 + 3xy^2 + g(y) \Rightarrow v_y^{'} = 6xy + g'(y) \\ v_y^{'} = u_x^{'} = y+6xy \end{cases}$$

Thus we have that

$$g'(x) = y \Rightarrow g(y) = \frac{y^2}{2} + C,$$

thus

$$v(x,y) = \frac{y^2}{2} + 3xy^2 - \frac{x^2}{2} - x^3 + C.$$

In paranthesis we remark that the analytic function $f(z) = u + iv$ is given by

$$f(z) = xy+3x^2y-y^3+i\left(\frac{y^2}{2} + 3xy^2 - \frac{x^2}{2} - x^3 + C\right) = -iz^2(z+1/2)+iC.$$

(b) Set

$$u(x,y)=e^{x^2-y^2}\cos(2xy)\Rightarrow\begin{cases}u_x^{'}=2xe^{x^2-y^2}\cos(2xy)-2ye^{x^2-y^2}\sin(2xy)\\ u_y^{'}=-2ye^{x^2-y^2}\cos(2xy)-2xe^{x^2-y^2}\sin(2xy)\end{cases}\Rightarrow$$

$$\Rightarrow\begin{cases}u_{xx}^{''}=2e^{x^2-y^2}(\cos 2xy+2x^2\cos 2xy-2y^2\cos 2xy-4xy\sin 2xy)\\ u_{yy}^{'}=-2e^{x^2-y^2}(\cos 2xy+2x^2\cos 2xy-2y^2\cos 2xy-4xy\sin 2xy)\end{cases}\Rightarrow\triangle u=0,$$

thus the function $u(x,y)$ is harmonic. Cauchy Riemann give us

$$\begin{cases}v_x^{'}=-u_y^{'}=2ye^{x^2-y^2}\cos(2xy)+2xe^{x^2-y^2}\sin(2xy)\Rightarrow v(x,y)=e^{x^2-y^2}\sin(2xy)+g(y)\Rightarrow\\ \Rightarrow v_y^{'}=-2ye^{x^2-y^2}\sin(2xy)+2xe^{x^2-y^2}\cos(2xy)+g'(y)\\ v_y^{'}=u_x^{'}=2xe^{x^2-y^2}\cos(2xy)-2ye^{x^2-y^2}\sin(2xy)\end{cases}$$

Thus we have that

$$g'(y)=0\Rightarrow g(y)=C,$$

thus

$$v(x,y)=e^{x^2-y^2}\sin(2xy)+C$$

In paranthesis we remark that the analytic function $f(z)=u+iv$ is given by

$$f(z)=e^{x^2-y^2}\cos(2xy)+i\left(e^{x^2-y^2}\sin(2xy)+C\right)=e^{z^2}+iC.$$

(c) Set

$$u(x,y)=\tan^{-1}(y/x),x>0\Rightarrow$$
$$\Longrightarrow\begin{cases}u_x^{'}=\dfrac{1}{1+(y/x)^2}\left(-\dfrac{y}{x^2}\right)=\dfrac{-y}{x^2+y^2}\\ u_y^{'}=\dfrac{x}{x^2+y^2}\end{cases}\Longrightarrow$$
$$\Rightarrow\begin{cases}u_{xx}^{''}=\dfrac{2xy}{(x^2+y^2)^2}\\ u_{yy}^{'}=\dfrac{-2xy}{(x^2+y^2)^2}\end{cases}\Rightarrow\triangle u=0,$$

thus the function $u(x,y)$ is harmonic. Cauchy Riemann give us

$$\begin{cases}v_x^{'}=-u_y^{'}=\dfrac{-x}{x^2+y^2}\Rightarrow v(x,y)=a-\frac{1}{2}\log(x^2+y^2)+g(y)\Rightarrow v_y^{'}=\dfrac{-y}{x^2+y^2}+g'(y)\\ v_y^{'}=u_x^{'}=\dfrac{-y}{x^2+y^2}\end{cases}$$

Thus we have that

$$g'(y)=0\Rightarrow g(y)=C,$$

thus

$$v(x,y)=-\frac{1}{2}\log\left(x^2+y^2\right)+C.$$

(d) Set

$$u(x,y)=\frac{x}{x^2+y^2}\Rightarrow\begin{cases}u_x^{'}=-\dfrac{x^2-y^2}{(x^2+y^2)^2}\\ u_y^{'}=-\dfrac{2xy}{(x^2+y^2)^2}\end{cases}\Rightarrow\begin{cases}u_{xx}^{''}=-2x\dfrac{3y^2-x^2}{(x^2+y^2)^3}\\ u_{yy}^{'}=2x\dfrac{3y^2-x^2}{(x^2+y^2)^3}\end{cases}\Rightarrow\triangle u=0,$$

thus is the function $u(x,y)$ is harmonic. Cauchy Riemann give us

$$\begin{cases} v_x^{'} = -u_y^{'} = \dfrac{2xy}{\left(x^2+y^2\right)^2} \Rightarrow v\left(x,y\right) = \dfrac{-y}{\left(x^2+y^2\right)} + g\left(y\right) \Rightarrow v_y^{'} = -\dfrac{x^2-y^2}{\left(x^2+y^2\right)^2} + g^{'}\left(y\right) \\ v_y^{'} = u_x^{'} = -\dfrac{x^2-y^2}{\left(x^2+y^2\right)^2} \end{cases}$$

Thus we have that

$$g^{'}\left(y\right) = 0 \Rightarrow g\left(y\right) = C,$$

thus

$$v\left(x,y\right) = \frac{-y}{\left(x^2+y^2\right)} + C$$

In paranthesis we remark that the analytic function $f\left(z\right) = u + iv$ is given by

$$f\left(z\right) = \frac{x}{x^2+y^2} + i\left(\frac{-y}{\left(x^2+y^2\right)} + C\right) = \frac{\bar{z}}{\left|z\right|^2} + iC = \frac{\bar{z}}{z\bar{z}} + iC = \frac{1}{z} + iC.$$